Rayyan Hussain

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EDUCATION

Georgia Institute of Technology **Expected Graduation: May 2028**
B.S. Computer Engineering

WORK EXPERIENCE

CrowdStrike | Software Engineer Intern | *Sunnyvale, CA* **Jun 2026 – Sep 2026**
- Designed and built LLM-powered agents automating 3 stages of an internal development pipeline (code review triage, build validation, deploy checks), eliminating a manual step performed on every PR.
- Developed API-driven automation to reconcile 10K+ configuration items (CMDB) across 30+ services to cut config drift and manual data upkeep.
- Integrated agentic workflows into existing CI/CD tooling, targeting a projected ~6 hours/week reduction in engineering toil for the platform team.

Stealth Startup | Co-Founder / Lead Engineer | *San Francisco, CA* **Jan 2026 – Present**
- Building a real-time AI **voice agent** to empower **developer productivity**.
- Implementing the agent with continuous listening, combining **speech + text** into LLM context, and streaming inference with early-start TTS and a **Redis** "sentence-starter" cache to enable natural turn-taking at **~50–100 ms perceived latency**, with **PostgreSQL** for durable session state and analytics.
- Created a workflow for users to write code and test execution, integrating responses for natural input handling.

Robotics Systems Lab @ SCU | Software Engineer Intern | *Santa Clara, CA* **Jun 2025 – Dec 2025**
- Co-authored paper on 3-D cluster control; published for the **2026 IEEE Aerospace Conference**.
- Awarded **Kuehler Grant** to research **multirobot adaptive-navigation** with **cluster-controlled drones**.
- Implemented PID loops in ROS 2 that kept a two-drone **autonomous** formation within 0.1 m of desired paths.
- Designed a 24-camera OptiTrack testbed (0.2 mm accuracy) for real-time 3-D tracking of Crazyflie microdrones.
- Built modular **Linux**-based distributed systems enabling real-time 3D tracking and control of Crazyflie microdrones.

PROJECTS

Power2ThePeople | NVIDIA Spark Hack Series – San Francisco **Jan 2026**
- Won **1st place** at **NVIDIA** hackathon (January 23–25, 2026) for building a real-time, situationally aware legal guidance platform.
- Utilized the NVIDIA DGX Spark to locally run the **Nemotron-70B model** with a visual language model, speech recognition input from an iOS app built in **Swift**, the **NV-Embed-v2 embedding** model for California and SF traffic laws, and geographic data from SF open datasets.
- Reduced latency compared to conventional cloud models by 20x.

DevAngel | AWS x INRIX Hackathon 2025 **Oct 2025**
- Won **2nd place** at **AWS Hackathon** for building an end-to-end automated incident response platform that detects production errors through **AWS CloudWatch**, analyzes root causes, generates and submits GitHub pull requests with fixes, and alerts support teams reducing mean time to recovery by **10x**.
- Developed a full-stack solution using **React** for the frontend, **AWS Lambda** and Step Functions for serverless backend automation, and **Amazon S3** for storing historical incident snapshots with CI/CD deployment on **EC2**.
- Integrated **Amazon Q** to automatically parse error logs, identify root causes, and generate **GitHub** pull requests for fixes, creating a fully autonomous recovery workflow that eliminates manual log parsing and reduces engineer burnout.

Sheltr | SCU Hack For Humanity 2025 **Feb 2025**
- Won **4th place** at Hack For Humanity among 350+ participants; won **Best Freshman Hack** for developing a disaster relief platform helping individuals find nearby verified shelters during emergencies.
- Built a full-stack web app using **FastAPI (Python)**, **AWS DynamoDB**, AWS EC2 (hosted via NginX), and Google Maps APIs, enabling location-based shelter recommendations and real-time reservation management.

LEADERSHIP

Association of Computing Machinery (ACM) | Vice President | *Santa Clara, CA* **May 2025 – Present**
- Leading end-to-end planning and execution of campus-wide hackathons at Santa Clara University, drawing 1,000+ participants; raised $85,000 in annual club funds within 4 months of joining the board.
- Coordinating with engineers and managers from **AWS**, **Intel**, **ServiceNow**, and **Roblox** through structured sprint planning, tracking milestones, and ensuring deliverables stay on schedule.

SKILLS

Technical: Python, Java, C, C++, Swift, Git, HTML, React, SQL, Postman, Go, Flask, AWS, ROS, Linux, Docker
Functional: Creative problem solving, logic, rhetoric, evidence-based decisions, user-centered design, effective communication